

# 4339 Asset Pricing and Investments - Assignment 1

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## 1 Covariance:

The returns for two risky assets A and B are given for three equally likely states

| Returns | State 1 | State 2 | State 3 |
|---------|---------|---------|---------|
| Asset 1 | -2%     | -1%     | 3%      |
| Asset 2 | 5%      | 3%      | -5%     |

### 1.1 Expected Returns, standard deviation and correlation coefficient

The expected return  $\mathbb{E}[r]$  is given by

$$\mathbb{E}[r] = \sum_{s=1}^S p_s r_s$$

where  $S$  is the total number of states and  $p_s$  is the probability of state  $s$ . Since all three states are equally likely,  $p_s = \frac{1}{S}$  for all  $s$ , this simplifies to:

$$\mathbb{E}[r] = \frac{1}{S} \sum_{s=1}^S r_s$$

Substituting the values from the table yields

$$\mathbb{E}[R_A] = \frac{1}{3}(-2 - 1 + 3) = 0\%$$

$$\mathbb{E}[R_B] = \frac{1}{3}(5 + 3 - 5) = 1\%$$

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The variance of returns is given by

$$\text{Var}(r) = \sigma^2 = \sum_{s=1}^S p_s (r_s - \mathbb{E}[r])^2$$

with the standard deviation being its square root

$$\sigma = \sqrt{\text{Var}(r)} = \sqrt{\sum_{s=1}^S p_s (r_s - \mathbb{E}[r])^2}$$

As above,  $p_s$  can be pulled out of the sum as the common factor.

Substituting the values from the table yields

$$\sigma_A = \sqrt{\frac{1}{3}[(-2-0)^2 + (-1-0)^2 + (3-0)^2]} = \sqrt{\frac{14}{3}} \approx 2.16\%,$$

$$\sigma_B = \sqrt{\frac{1}{3}[(5-1)^2 + (3-1)^2 + (-5-1)^2]} = \sqrt{\frac{56}{3}} \approx 4.32\%$$

The correlation coefficient is given by

$$\rho_{ij} = \text{Corr}[r_i, r_j] = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

where the covariance  $\text{Corr}[r_i, r_j]$  is defined as

$$\text{Cov}[r_i, r_j] = \mathbb{E}[(r_i - \mathbb{E}[r_i])(r_j - \mathbb{E}[r_j])]$$

Writing  $\mu_i = \mathbb{E}[r_i]$  and  $\mu_j = \mathbb{E}[r_j]$ , which are constants, expanding the product and applying linearity of expectations gives

$$\begin{aligned} \sigma_{ij} &= \mathbb{E}[r_i r_j - \mu_j r_i - \mu_i r_j + \mu_i \mu_j] \\ &= \mathbb{E}[r_i r_j] - \mu_j \mathbb{E}[r_i] - \mu_i \mathbb{E}[r_j] + \mu_i \mu_j \\ &= \mathbb{E}[r_i r_j] - \mu_i \mu_j \\ &= \mathbb{E}[r_i r_j] - \mathbb{E}[r_i] \mathbb{E}[r_j] \end{aligned}$$

Substituting the values from the table yields

$$\sigma_{AB} = \frac{1}{3}[(-2)(5) + (-1)(3) + (3)(-5)] - (0)(1) = -\frac{28}{3} \approx -9.33 \text{ (\%}^2\text{)}.$$

$$\rho_{AB} = \frac{-\frac{28}{3}}{\sqrt{\frac{14}{3}} \cdot \sqrt{\frac{56}{3}}} = \frac{-\frac{28}{3}}{\sqrt{\frac{784}{9}}} = \frac{-\frac{28}{3}}{\frac{28}{3}} = -1$$

The assets are perfectly negatively correlated.

## 1.2 The fully invested Portfolio, made of A and B, with minimum variance

Since the portfolio is fully invested it follows that  $w_B = 1 - w_A$  and the portfolio variance can be written as

$$\sigma_p^2 = w_A^2 \sigma_A^2 + (1 - w_A)^2 \sigma_B^2 + 2w_A(1 - w_A)\sigma_{AB}$$

To find the minimum Variance Portfolio we differentiate the variance function with respect to  $w_A$ ,

$$\frac{\partial \sigma_p^2}{\partial w_A} = 2w_A \sigma_A^2 - 2(1 - w_A) \sigma_B^2 + (2 - 4w_A) \sigma_{AB}$$

and set the derivative equal to zero

$$\begin{aligned} 0 &= 2w_A \sigma_A^2 - 2(1 - w_A) \sigma_B^2 + (2 - 4w_A) \sigma_{AB} \\ 0 &= w_A \sigma_A^2 - \sigma_B^2 + w_A \sigma_B^2 + \sigma_{AB} - 2w_A \sigma_{AB} \\ 0 &= w_A (\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}) - \sigma_B^2 + \sigma_{AB} \\ w_A^* &= \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}} \end{aligned}$$

Substituting  $\sigma_A^2 = \frac{14}{3}$ ,  $\sigma_B^2 = \frac{56}{3}$  and  $\sigma_{AB} = -\frac{28}{3}$

$$w_A^* = \frac{\frac{56}{3} + \frac{28}{3}}{\frac{14}{3} + \frac{56}{3} + \frac{56}{3}} = \frac{28}{42} = \frac{2}{3}, \quad w_B^* = \frac{1}{3}.$$

The resulting portfolio variance is

$$\sigma_p^2 = \frac{4}{9} \cdot \frac{14}{3} + \frac{1}{9} \cdot \frac{56}{3} + 2 \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot \left(-\frac{28}{3}\right) = \frac{56}{27} + \frac{56}{27} - \frac{112}{27} = 0.$$


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## 1.3 The set of all possible arbitrage portfolios when a risk-free bond with return 0% exists and a no-short sale constraint is imposed on A and B

The zero-cost portfolio therefore borrows at  $r_f = 0\%$  and invests the money in the two risky assets. To begin with, we normalise the borrowed amount to one unit and

$$w_A + w_B = 1, \quad w_A, w_B \geq 0,$$

so that  $w_B = 1 - w_A$  with  $w_A \in [0, 1]$ .

As the bonds return is 0, the portfolio return in state  $s$  is  $r_{P,s} = w_A r_{A,s} + (1 - w_A) r_{B,s}$ . Following the no-arbitrage condition this must yield a non-negative amount in every state, which allows us to set up three inequalities

$$\text{State 1: } 0 \leq -2w_A + 5(1 - w_A) = 5 - 7w_A \implies w_A \leq \frac{5}{7} \approx 0.7143,$$

$$\text{State 2: } 0 \leq -1w_A + 3(1 - w_A) = 3 - 4w_A \implies w_A \leq \frac{3}{4} = 0.75,$$

$$\text{State 3: } 0 \leq 3w_A - 5(1 - w_A) = 8w_A - 5 \implies w_A \geq \frac{5}{8} = 0.625$$

Since the first condition is stricter than the second one, only conditions (1) and (3) are binding. Combining the remaining two yields

$$\frac{5}{8} \leq w_A \leq \frac{5}{7} \quad \text{i.e.} \quad 0.625 \leq w_A \leq 0.7143$$

As there exists no  $w_A$  which satisfies all three conditions with equality simultaneously, i.e.  $w_A = \frac{5}{7}$ ,  $w_A = \frac{3}{4}$  and  $w_A = \frac{5}{8}$ , each portfolio earns a strictly positive return in at least one state.

By normalising to a single unit of borrowed money, we obtained the proportions of  $w_A$  and  $w_B$ . Scaling the position up or down by shorting more or less of the risk-free asset gives all possible arbitrage portfolios  $(w_A, w_B, w_f)$  as  $x \cdot (w_A, 1 - w_A, -1)$  for any  $x > 0$  and any  $w_A \in [\frac{5}{8}, \frac{5}{7}]$ .

## 2 Optimal Portfolio Allocation with two risky assets:

The expected return, standard deviation and the covariance of a corporate bond fund and an equity fund are given

| Security   | Expected Return | Standard Deviation | Covariance |
|------------|-----------------|--------------------|------------|
| Stock Fund | 10%             | 20%                | 0.0002     |
| Bond Fund  | 5%              | 8%                 |            |

### 2.1 Compute and plot the efficient frontier

To solve this we use `opt_portfolio_2asset.ipynb` with the values referring to the table above.

```
[33]: import numpy as np
import matplotlib.pyplot as plt

# Setup
# Basic inputs for bonds and stocks

# Mean
SB_mean = np.array([[0.1], [0.05]])

# Variance-Covariance Matrix
SB_Sigma = np.array([[0.2**2, 0.0002], [0.0002, 0.08**2]])

# Vector of ones
ones2 = np.ones((2, 1))
```

```
# Inverse of the variance-covariance matrix
SB_invSigma = np.linalg.inv(SB_Sigma)
```

```
[34]: # Define vectors for weights: Stock = (0,1)^t and Bond = (1,0)^t
E_w = np.array([[1], [0]]) # Bond
B_w = np.array([[0], [1]]) # Stock

# Create many portfolios as a combination of stock and bond

# Possible combinations of weights
y = np.arange(-1, 2.01, 0.01)

# Convert y to a 2D array where each row is a weight for a portfolio
y = y.reshape(1, -1)
port = np.dot(B_w, y) + np.dot(E_w, (1 - y))

# Compute expected returns
port_mean = np.dot(port.T, SB_mean)

# Compute var-covar matrix of portfolio returns (we will only use the variances
↳ on the main diagonal)
port_var = np.diag(np.dot(np.dot(port.T, SB_Sigma), port))
port_var = np.clip(port_var, 0, None) # guard against tiny negative values
↳ from float rounding
port_vol = np.sqrt(port_var).reshape(len(port_var), 1)
```

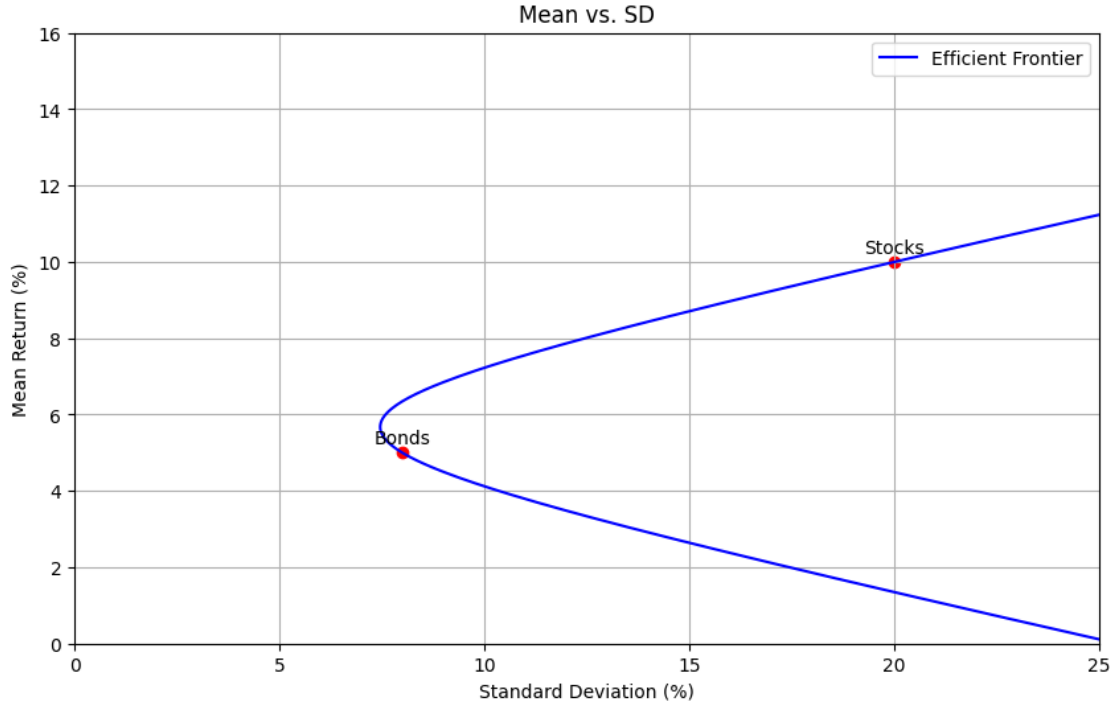
```
[35]: # Plot MV
plt.figure(figsize=(10, 6))
plt.plot(port_vol * 100, port_mean * 100, 'b-', label='Efficient Frontier')
plt.title("Mean vs. SD")
plt.xlabel("Standard Deviation (%)")
plt.ylabel("Mean Return (%)")
plt.xlim(0, 25)
plt.ylim(0, 16)

# Add points for stock and bond funds
plt.scatter(np.sqrt(np.diag(SB_Sigma)) * 100, SB_mean * 100, color='red')
SB_labels = ["Stocks", "Bonds"]

# Annotate points
for i, txt in enumerate(SB_labels):
    plt.annotate(txt, (np.sqrt(np.diag(SB_Sigma))[i] * 100, SB_mean[i, 0] *
↳ 100), textcoords="offset points", xytext=(0, 5), ha='center')

plt.legend()
plt.grid(True)
```

```
plt.show()
```



## 2.2 Compute the asset weights that make up the minimum variance portfolio (GMV)

Following *Section 1.3* the weights of the fully invested global minimum variance portfolio, with 2 risky assets, are

$$w_A^* = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}} \quad \text{and} \quad w_B^* = 1 - w_A^*$$

Substituting  $\sigma_A^2 = 0.2^2$ ,  $\sigma_B^2 = 0.08^2$  and  $\sigma_{AB} = 0.0002$

$$w_A^* = \frac{0.08^2 - 0.0002}{0.2^2 + 0.08^2 - 0.0004} = \frac{31}{230} \approx 0.1349, \quad w_B^* = \frac{199}{230} \approx 0.8652.$$

The resulting portfolio variance is

$$\sigma_P^2 = \left(\frac{31}{230}\right)^2 0.04 + \left(\frac{199}{230}\right)^2 0.0064 + 2 \cdot \frac{31}{230} \cdot \frac{199}{230} \cdot 0.0002 \approx 0.005564$$

and the standard deviation is

$$\sigma_P \approx 0.074595 = 7.46\%$$

The expected return of the GMV portfolio is

$$\mu_P = \left(\frac{31}{230}\right) 0.10 + \left(\frac{199}{230}\right) 0.05 \approx 0.0567 = 5.67\%.$$

It can be seen that holding a certain amount of the riskier stock fund reduces the overall portfolio risk compared to the safer bond fund,  $\sigma_P = 7.46\% < \sigma_B = 8\%$ .

### 2.3 Assuming a risk-free asset, with $r_f = 0.03$ , derive the tangent portfolio and the optimal CAL. Plot them on a graph with the efficient frontier.

The tangent portfolio maximizes the Sharpe Ratio  $S$ , which is defined as

$$S = \frac{\mathbb{E}[r_T] - r_f}{\sigma_T}$$

For the case with two risky assets let  $g(w_A)$  be the excess return function and  $h(w_A)$  be the variance function

$$\begin{aligned} g &= w_A r_A + (1 - w_A) r_B - r_f \\ h &= w_A^2 \sigma_A^2 + (1 - w_A)^2 \sigma_B^2 + 2w_A(1 - w_A) \sigma_{AB} \end{aligned}$$

As the tangency portfolio lies on the efficient frontier and is as such made up of only the two risky assets, we can use  $w_B = 1 - w_A$ .

To maximize the Sharpe Ratio we differentiate  $S$  with respect to  $w_A$  and set this FOC equal to 0

$$\begin{aligned} S(w_A) &= \frac{g}{\sqrt{h}} \\ S(w_A) &= g h^{-1/2} \end{aligned}$$

By applying the product rule we find the FOC

$$\begin{aligned} \frac{\partial S}{\partial w_A} &= g' h^{-1/2} + g \left(-\frac{1}{2}\right) h^{-3/2} h' \\ &= h^{-3/2} [g' h - \frac{1}{2} g h'] \end{aligned}$$

Now we can set the derivative equal to 0 and multiply by  $h^{3/2}$ , which gives

$$\begin{aligned} g' h - \frac{1}{2} g h' &= 0 \\ 2g' h &= g h' \end{aligned}$$

Substituting into  $2g' h = g h'$ , differentiating the remaining parts and simplifying, results in

$$w_A^T = \frac{(r_A - r_f) \sigma_B^2 - (r_B - r_f) \sigma_{AB}}{(r_A - r_f) \sigma_B^2 + (r_B - r_f) \sigma_A^2 - [(r_A - r_f) + (r_B - r_f)] \sigma_{AB}}$$

Substituting  $r_A = 0.10$ ,  $r_B = 0.05$ ,  $r_f = 0.03$ ,  $\sigma_A^2 = 0.2^2$ ,  $\sigma_B^2 = 0.08^2$  and  $\sigma_{AB} = 0.0002$

$$w_A^T = \frac{(0.07)0.0064 - (0.02)0.0002}{(0.07)0.0064 + (0.02)0.04 - (0.09)0.0002} = \frac{74}{205} \approx 0.3610, \quad w_B^T = \frac{131}{205} \approx 0.6390.$$

The optimal CAL is the linear function through the risk-free asset and the tangency portfolio  $T$  with  $r_f$  as the y-intercept and the Sharpe Ratio as the slope

$$CAL = r_f + \frac{E[r_T] - r_f}{\sigma_T} \sigma_{CAL} = 0.03 + 0.4276 \sigma_C,$$


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The following code to generate the graph was produced by Claude.

```
[36]: # Tangency portfolio (risk-free rate 3%)
rf = 0.03

# Unpack: A = stock fund (index 0), B = bond fund (index 1)
rA, rB = SB_mean[0, 0], SB_mean[1, 0]
sA2, sB2 = SB_Sigma[0, 0], SB_Sigma[1, 1]
sAB = SB_Sigma[0, 1]

# Shorthand as in the derivation
m = rA - rB
c = rB - rf
a = sA2 + sB2 - 2 * sAB
b = sB2 - sAB

# FOC solution: w_A = (m*sB2 + b*c) / (m*b + a*c)
w_A_tan = (m * sB2 + b * c) / (m * b + a * c)
w_B_tan = 1 - w_A_tan

# Moments of the tangency portfolio
mu_tan = w_A_tan * rA + w_B_tan * rB
sd_tan = np.sqrt(w_A_tan**2 * sA2 + w_B_tan**2 * sB2 + 2 * w_A_tan * w_B_tan * sAB)
sharpe = (mu_tan - rf) / sd_tan

# Plot MV
plt.figure(figsize=(10, 6))
plt.plot(port_vol * 100, port_mean * 100, 'b-', label='Efficient Frontier')

# CAL and tangency portfolio      <-- add here, same cell
sd_grid = np.linspace(0, 0.25, 100)
plt.plot(sd_grid * 100, (rf + sharpe * sd_grid) * 100, color='orange',
         label=f'CAL (slope = {sharpe:.4f})')
plt.scatter(sd_tan * 100, mu_tan * 100, color='black', marker='D', zorder=5)
plt.annotate("Tangency", (sd_tan * 100, mu_tan * 100),
            textcoords="offset points", xytext=(0, 8), ha='center')

plt.title("Mean vs. SD")
plt.xlabel("Standard Deviation (%)")
```

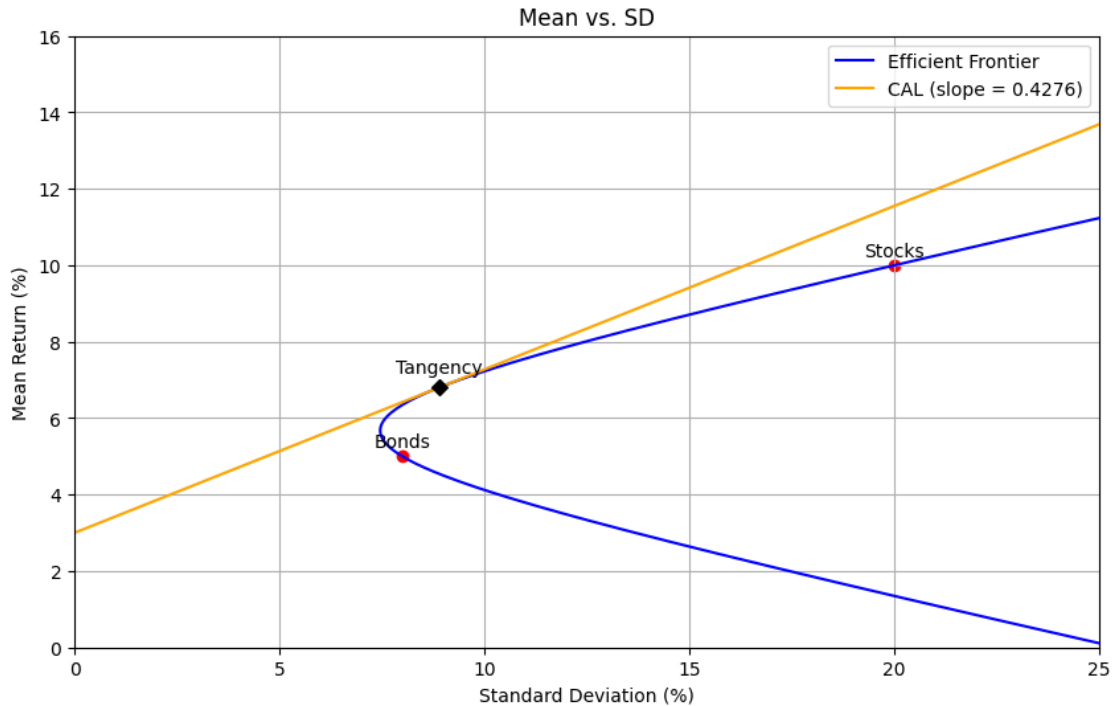
```

plt.ylabel("Mean Return (%)")
plt.xlim(0, 25)
plt.ylim(0, 16)

plt.scatter(np.sqrt(np.diag(SB_Sigma)) * 100, SB_mean * 100, color='red')
SB_labels = ["Stocks", "Bonds"]
for i, txt in enumerate(SB_labels):
    plt.annotate(txt, (np.sqrt(np.diag(SB_Sigma))[i] * 100, SB_mean[i, 0] * 100),
                 textcoords="offset points", xytext=(0, 5), ha='center')

plt.legend()
plt.grid(True)
plt.show()

```



## 2.4 The optimal portfolio for an investor with risk aversion parameter $a = 4$ and its certainty equivalent

The investor can now move along the CAL by splitting his investment between the risky tangency portfolio and the risk-free asset. Therefore, a risk averse investor maximizes his utility according to the following utility function

$$U^* = r_f + w^*(\mathbb{E}[r_T] - r_f) - \frac{a}{2}(w^*)^2\sigma_T^2$$

To find the maximum we differentiate with respect to  $w$  and set it equal to 0

$$\frac{dU}{dw} = (\mathbb{E}[r_T] - r_f) - a w \sigma_T^2 = 0 \quad \implies \quad w^* = \frac{(\mathbb{E}[r_T] - r_f)}{a \sigma_T^2}$$

Substituting the values yields

$$w^* \approx \frac{0.06805 - 0.03}{4(0.007918)} = 1.2014$$

To maximize his utility the investor takes on leverage by shortselling 20.14% of the risk-free asset and investing 120.14% into the tangency portfolio.

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The certainty equivalent is the risk-free rate  $r_f^*$  that provides the investor with the same utility as the above portfolio. As a sole investment in a risk-free asset carries no risk (volatility), the utility functions simplifies to

$$U^* = r_f^*$$

To find this rate we set the “risk-free utility function” equal to the general utility function from above

$$\begin{aligned} r_f^* &= r_f + w^*(\mathbb{E}[r_T] - r_f) - \frac{a}{2}(w^*)^2 \sigma_T^2 \\ r_f^* &\approx 0.03 + 1.2014(0.06805 - 0.03) - \frac{4}{2}(1.2014)^2(0.007918) \\ r_f^* &\approx 0.05286 \end{aligned}$$

The investor is thus indifferent between his optimal risky portfolio and a safe return of 5.29%

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## 2.5 Effects of changes to the risk aversion parameter $a$ and to the risk-free rate $r_f$

As shown in Section 2.3, the tangency portfolio is not defined by the risk aversion parameter. Therefore increasing  $a$  to 8 has no effect on the stock-bond split that makes up the tangency portfolio.

However, it affects the position the investor takes along the CAL in order to maximise his utility. Following the optimal weight formula from Section 2.4,  $w^*$  is inversely proportional to  $a$ , and since neither  $\mathbb{E}[r_T]$  nor  $\sigma_T$  depends on risk aversion, doubling  $a$  exactly halves the weight placed in the tangency portfolio

$$a = 2 \cdot 4 = 8 \quad \implies \quad w^* = \frac{1}{2}(1.2014) = 0.6007.$$

This is also known as the two fund separation theorem.

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In contrast, changing the risk-free rate to  $r_f = 0.04$  affects the Sharpe maximisation, which defines the tangency portfolio. The weight in stocks  $w_A$  rises from 36.1% to 49.6%, moving the tangency portfolio to nearly an even split between stock and bond.

This is because, while in absolute terms both assets lose an equal 1% of excess return, the bond loses more in relative terms. Its premium is halved from 2% to 1%, whereas the equity premium falls only from 7% to 6%. As the volatilities remain unchanged, the bond loses more of its return-to-risk ratio. Consequently the portfolio Sharpe maximisation favours the stock portion more.

### 3 Mean-Variance Frontier:

The following 6 risky assets are available to trade

|                        | Asset 1 | Asset 2 | Asset 3 | Asset 4 | Asset 5 | Asset 6 |
|------------------------|---------|---------|---------|---------|---------|---------|
| Expected Return (%)    | 8%      | 10%     | 9%      | 5%      | 12%     | 15%     |
| Standard Deviation (%) | 10%     | 12%     | 16%     | 7%      | 15%     | 20%     |

#### 3.1 One asset that is dominated by another asset

An asset is dominated by another asset if it provides a lower expected return while having an equal or higher amount of risk or if it provides a lower or equal expected return while having an higher amount of risk.

This is the case for asset 3 and asset 2 where both inequalities hold strictly

$$\mathbb{E}[r_3] = 9\% < 10\% = \mathbb{E}[r_2] \quad \text{and} \quad \sigma_3 = 16\% > 12\% = \sigma_2$$

As such, asset 2 dominates asset 3 by offering a higher expected return while carrying a lower amount of risk. It should be noted, that in portfolio allocation this does not necessarily result in the exclusion of asset 3, as the covariance between the assets is also a critical measure.

#### 3.2 Assets that lie on the efficient frontier given a no-short sale constraint

Following the no-short sale constraint, the expected portfolio return for  $N$  assets, which are indexed by  $i$ , is

$$\mathbb{E}[r_P] = \sum_{i=1}^N w_i r_i \quad \sum_{i=1}^N w_i = 1 \quad w_i \geq 0.$$

When a no-shortsale constraint is imposed, the highest achievable portfolio return is the portfolio that is fully invested in the highest returning asset.

$$\mathbb{E}[r_{P,max}] = 1 \cdot r_{i,max}$$

From the table above, the highest returning asset is asset 6.

The efficient frontier consists of all constructable portfolios that provide the highest level of return for a given level of risk. Since no other asset achieves a higher return and asset 6 is the only one achieving 15%, there exists no other portfolio that offers at least the same return for a lower or equal amount of risk. Asset 6 must therefore lie on the efficient frontier regardless of its volatility of 20%.